Continuous-time signals I.

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Please open Python notebooks **`continuous_1, fs, ft`**

Agenda

- **Basic information, operations with time**
- Important signals – with links to discrete ones
- Periodic signals
- Energy and power
- Spectral analysis of periodic signals – Fourier series
- FS of typical signals
- Properties of FS and power by FS
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Remember lecture 1 … functions and sequences

- Signals with **continuous time** (analog, real world) are defined for all times *t*. Signály se spojitým časem, spojité signály.
  - From math point of view, they are **functions** $x(t)$ ← round brackets
- Signals with **discrete time** (computer, digital world), sampled signals are defined only for integer times (sample numbers) *n*. Signály s diskrétním časem, diskrétní signály, vzorkované signály.
  - From math point of view, they are **sequences** $x[n]$ ← square brackets.

Continuous time signals

- Time (horizontal axis) is in seconds [s]
- Frequencies will be true ones - in 1/s – Hertz [Hz]
- Continuous−time signals are really defined for **all** times.
- When plotting, we'll use some small time-step and we'll definitely use `plot`.
- Reminder – for discrete-time signals, we either emphasized that they are discrete and used `stem`, or did not care and used `plot`.

#continuous_plotting

- When generating a signal in Python (or any other programming language), you can advantageously use indexing by results of conditions.

#use_conditions

Time modifications of continuous signals

- Reversal of the time axis (otočení): $y(t) = x(-t)$
- Basic time shifts
  - Delay (zpoždění) – shift to the right: $y(t) = x(t - \tau)$
  - Advance (předběhnutí) – shift to the left: $y(t) = x(t + \tau)$
- Reversal of the time axis and time shifts:
  - Reversal and shift to the right: $y(t) = x(-t + \tau)$
  - Reversal and shift to the left: $y(t) = x(-t - \tau)$
- Unsure ? Run a check:
  - Select an important event in the signal
  - For the modified signal, evaluate the time modification for this event
  - Look at the resulting time in the original signal.
    - Same event as in the modified one ? **OK**
    - Different event ? **BAD, DO IT AGAIN**

#time_modifications

Modifying the speed of time …

- Contraction of time (kontrakce, zrychlení času): $y(t) = x(mt)$ for $m > 1$
  – everything will run faster
- Dilation of time (dilatace, zpomalení času): $y(t) = x(mt)$ for $m < 1$
  – everything will run slower.
- We rarely manage to slow-down or speed-up time in real world, but we can use this know-how to compute / draw a faster or slower signal provided we know the original one.

#time_contraction_dilation

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Constant signal

- Discrete: $x[n] = const$
- Continuous: $x(t) = const$
- Constant signals are very useful – remember charging your cell phone ☺
- Constant signal has zero frequency.

#constant_signal

Unit step (jednotkový skok) and unit pulse (jednotkový impuls) – discrete

- Unit step: all samples below *n = 0* are zero:

$$\sigma[n] = \begin{cases} 1 & \text{for} \quad n \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

- Unit pulse: all samples except *n = 0* are zero:

$$\delta[n] = \begin{cases} 1 & \text{for} \quad n = 0 \\ 0 & \text{elsewhere} \end{cases}$$

- Unit pulse can be given as a **difference** of basic and shifted unit steps:
  $\delta[n] = \sigma[n] - \sigma[n-1]$

#discrete_unit_step_pulse

Unit step in continuous time

- Unit step: signal below t = 0 is zero $\sigma(t) = \begin{cases} 1 & \text{for} \quad t \geq 0 \\ 0 & \text{elsewhere} \end{cases}$
- Unit pulse should be defined as $\delta(t) = \dfrac{d\sigma(t)}{dt}.$
- But it is hard to derive unit pulse, as the derivation in *t = 0* is not defined.
- Therefore, we define approximate (smooth) unit step with width *Δ* and derive it:

$$\sigma_\Delta(t) = \begin{cases} 0 & \text{for} \quad t < 0 \\ \frac{1}{\Delta}t & \text{for} \quad 0 \leq t \leq \Delta \\ 1 & \text{for} \quad t > 1 \end{cases}$$

#unit_step_derivation

Dirac pulse (Diracův impuls)

- The surface – “mighteness” (mocnost) is

$$M = \int_{-\infty}^{+\infty} \delta\Delta(t)dt = \int_0^{\Delta} \frac{1}{\Delta}dt = \frac{1}{\Delta}\Delta = 1.$$

- We run Δ to zero, and we obtain the Dirac pulse:

$$\delta(t) = \lim_{\Delta \to 0} \delta_\Delta(t).$$

  - The width is zero, the value is infinity.
  - The surface is however still one
- Mathematicians often refuse to call this a function, and rather call it distribution.
- We usually plot it with as a square with an arrow going to “infinity”.
- Dirac pulse can not be generated – it is a (very useful!) theoretical construct

#dirac

The sampling property of Dirac pulse

- Dirac is relatively aggressive when multiplied with signal *x(t):* it sets everything to zero, except *t = 0*.
- When integrated, the output value is *x(0):* the signal got sampled:

$$\int_{-\infty}^{+\infty} x(t)\delta(t)dt = x(0),$$

- In case we shift the Dirac pulse to time *τ*, we obtain similar behavior

$$\int_{-\infty}^{+\infty} x(t)\delta(t-\tau)dt = x(\tau).$$

- We will use this property when explaining the **sampling**.

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Periodic signals - continuous

- Periodic signals have repetitive behavior …
- We can find such *T*, that *x(t + T) = x(t)*
- *T* is called period, but several *T*'s can exist …
- T_1 is the smallest one, we call it **fundamental period** (základní perioda).
- Periodic signals have fundamental frequency (základní frekvence) in 1/s (Hertz $f_1 = \frac{1}{T_1}, \qquad T_1 = \frac{1}{f_1}.$
- We often work with angular/circular frequency/velocity (kruhová/úhlová frekvence/rychlost) $\omega_1 = 2\pi f_1 = \frac{2\pi}{T_1}, \qquad T_1 = \frac{2\pi}{\omega_1}.$

#periodic_continuous

Periodic signals - discrete

- We can find such N, that $x[n + N] = x[n]$,
- N is called period, but several N's can exist …
- N_1 is the smallest one: **fundamental period** (základní perioda).
- Periodic signals have fundamental normalized frequency (základní normovaná frekven $f_1 = \frac{1}{N_1}$]:
- We often work with normalized angular/circular frequency / velocity (normovaná kruhová/úhlová frekvence/rychlost $\omega_1 = \frac{2\pi}{N_1}$.
- Conversion from N_1 to f_1 and ω_1 is straightforward, but not vice versa.

#periodic_discrete

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Instantaneous power of a signal

- Remember Ohm's law:

$$p(t) = u(t)i(t) = u^2(t)/R = i^2(t)R.$$

- Therefore, we define **instantaneous power** (okamžitý výkon) for both discrete and continuous signals as:

$$p(t) = |x(t)|^2. \qquad p[n] = |x[n]|^2.$$

- Magnitude is not needed when we work with real signals, but we should not forget it for complex ones … and we do have complex signals in ISS !

Energy and powers of signals

- Energy (energie) is given as a sum (discrete signals) or integral (continuous signals) of instantaneous power over a time interv...

$$E_{n_2,n_2} = \sum_{n_1}^{n_2} p[n] = \sum_{n_1}^{n_2} |x[n]|^2. \qquad E_{t_2,t_2} = \int_{t_1}^{t_2} p(t)dt = \int_{t_1}^{t_2} |x(t)|^2 dt$$

- Mean power (... th of the time interval:

$$P_{t_2,t_2} = \frac{E_{t_1,t_2}}{t_2 - t_1} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} p(t)dt = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt$$

$$P_{n_2,n_2} = \frac{E_{n_1,n_2}}{n_2 - n_1 + 1} = \frac{1}{n_2 - n_1 + 1} \sum_{n_1}^{n_2} p[n] = \frac{1}{n_2 - n_1 + 1} \sum_{n_1}^{n_2} |x[n]|^2.$$

- Root mean square value (efektivní hodnota) ... $C_{ef} = \sqrt{P}.$ ...agnitude of a d.c. signal with equivalent power:

Powers of periodic signals and of a cosine

- When determining power of periodic signals, it is wise to compute over (and normalize by) one period: N_1 (discrete) and T_1 (continuous).
- Compu $s(t) = C_1 \cos(\omega_1 t)$ cosine is a basic task: T_1 and angular frequency ω_1.

$$\cos^2 \alpha = \frac{1+\cos 2\alpha}{2}$$

- With the help $P_s = \frac{1}{T_1}\int_0^{T_1} [C_1 \cos(\omega_1 t)]^2 dt = \frac{1}{T_1}\int_0^{T_1} {C_1}^2 \frac{1}{2}(1+\cos 2\omega_1 t) dt$ s the power as
- Th $P_s = \frac{1}{T_1}\int_0^{T_1} {C_1}^2 \frac{1}{2} dt = \frac{1}{T_1}\frac{{C_1}^2}{2} T_1 = \frac{{C_1}^2}{2}$ riods of cosir $C_{ef} = \sqrt{P_s} = \frac{C_1}{\sqrt{2}}$ power is:

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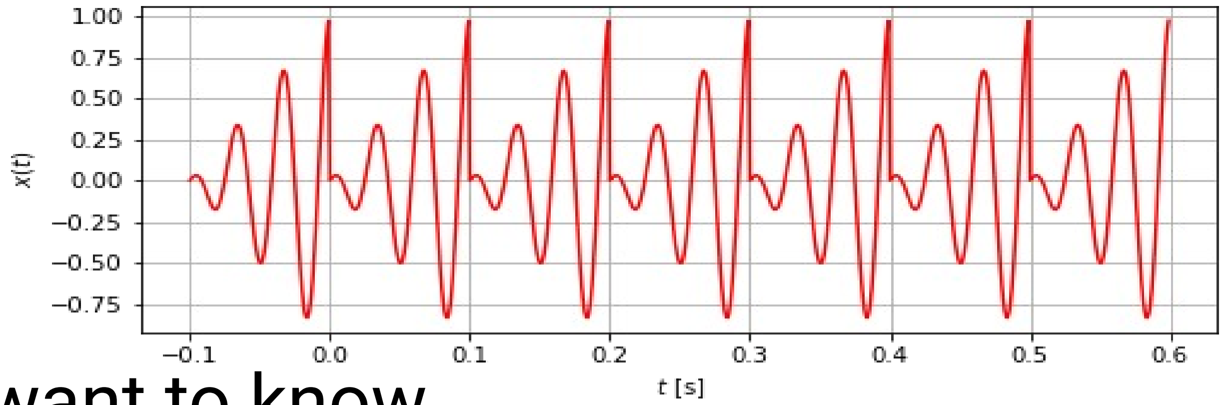
What do we require ?



- From any spectral analysis, we want to know
  - Where are the frequencies in the signal
  - How strong is the signal at a given frequency – “how much”
  - What is the phase of the signal at a given frequency – “how shifted”.
- We now have a periodic signal and we have seen that it can be composed of
  - The d.c. value
  - The slowest component (fundamental frequenc $\omega_1 = 2\pi f_1 = \frac{2\pi}{T_1}$.
  - Harmonically-related components at multiples of this frequency: $k\omega_1$
- We will define **Fourier series** (Fourierova řada)
  - It is series, not a function, we will work with **coefficients** !
  - They will “sit” on a given frequency.

Real formulation of FS

- We could write FS as a real function

$$x(t) = c_0 + C_1 \cos(1\omega_1 t + \phi_1) + C_2 \cos(2\omega_1 t + \phi_2) + C_3 \cos(3\omega_1 t + \phi_3) + \ldots$$

$$x(t) = c_0 + \sum_{k=1}^{\infty} C_k \cos(k\omega_1 t + \phi_k).$$

  - c_0 is D.C. component
  - C_k is magnitude at frequency $k\omega_1$
  - Φ_k is phase at frequency $k\omega_1$
- But this would make the computation of coefficients quite complicated
  - remember the problems with phase of cosine each time we tried to use them for analysis …

Complex formulation of FS

- Complex form of FS is more practical and more frequent:

$$x(t) = \ldots + c_{-3}e^{-j3\omega_1 t} + c_{-2}e^{-j2\omega_1 t} + c_{-1}e^{-j1\omega_1 t} + c_0 + c_1 e^{j1\omega_1 t} + c_2 e^{j2\omega_1 t} + c_3 e^{j3\omega_1 t} + \ldots$$

$$\boxed{x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_1 t}}$$

  - c_0 is D.C. component
  - c_k is a **complex** coefficient belonging to frequency $k\omega_1$
    - Its magnitude $|c_k|$ determines the "strength" of signal at given frequency
    - Its phase arg c_k determines the "shift" of signal at given frequency.
    - We need c_k at $k\omega_1$ and c_{-k} at $-k\omega_1$ to sum up to one real cosine – same trick as we used for DTFT.
- The formula above is called **synthesis formula**
- The values and positions of coefficients c_k are called **spectrum** (remember "spectrum" is a generic term).
- They are complex, so we will need 2 plots: magnitude and phase.

Relations between the real and complex forms

- The conversion is easily done by breaking cos into a sum of two complex exponentials:

$$x(t) = c_0 + \sum_{k=1}^{\infty} C_k \cos(k\omega_1 t + \phi_k) = c_0 + \sum_{k=1}^{\infty} \frac{C_k}{2} \left(e^{j(k\omega_1 t + \phi_k)} + e^{-j(k\omega_1 t - \phi_k)} \right).$$

- After re-arranging:

$$= c_0 + \sum_{k=1}^{\infty} \left(\frac{C_k}{2} e^{j\phi_k} e^{jk\omega_1 t} + \frac{C_k}{2} e^{-j\phi_k} e^{-jk\omega_1 t} \right) \quad = c_0 + \sum_{k=1}^{\infty} \frac{C_k}{2} e^{j\phi_k} e^{jk\omega_1 t} + \sum_{k=-\infty}^{k=-1} \frac{C_k}{2} e^{-j\phi_k} e^{jk\omega_1 t}$$

- So we see the conversion real -> complex and vice versa:

$$c_k = \frac{C_k}{2} e^{j\phi_k}, \quad c_{-k} = \frac{C_k}{2} e^{-j\phi_k} \qquad C_k = 2|c_k| = 2|c_{-k}|, \quad \phi_k = \arg c_k = -\arg c_{-k}.$$

- For real signals *x(t)*, there must be a symmetry of coefficients:

#cos_complex_exp

$$c_k = c^{\star}_{-k}$$

Computation of FS coefficients – **analysis formula**

- By FS, we are actually decomposing the signal (projecting, computing similarity with, computing correlation w $e^{jk\omega_1 t}$ to bases
- We know that the projection can be done by a scalar product of the signal and the analysis signal.
  - Discrete: $c = \sum x[n]a[n]$ Conti $c = \int x(t)a(t)dt$
- The signal is periodic, so it makes sense to integrate only over one period T_1 picked wherever we want:
  - 0 to T_1, or $-T_1/2$ to $T_1/2$ or …

$$\boxed{c_k = \frac{1}{T_1} \int_{T_1} x(t) e^{-jk\omega_1 t} dt}$$

- FS basis
  - is complex, so we need to take it in complex-conjugate form.
  - has norm T_1, (just try to compute the integral of its magnitude over one period T_1), so we need to divide by T_1.

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Examples of FS computation

- Examples
  - *x(t)* is complex exponential
  - *x(t)* is a cosine with magnitude and phase.
  - *x(t)* is a sequence of rectangular pulses
- Computing FS – always respect the rule of maximum laziness !
  - Looking at the definition formula – can't I find the FS coefficients directly ?
  - Integrating by hand (for "simple" deterministic signals, such as the rectangular pulses).
  - Integrating numerically – see next lecture, it will lead us to DFT.
- Exercise your **intuition**
  - Slow changing signal has little high frequencies – few non-zero FS coefficients, narrow spectrum
  - Fast changing signal has lots of high frequencies – lots of non-zero FS coefficients, wide spectrum
  - An edge or pulse in the signal is a very fast change !

FS of a complex exponential and cosine

$$\boxed{x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_1 t}}$$

- Plain complex exponentia $x(t) = e^{j\omega_1 t}$
  - This is the simples case, as we can directly relate it to the synthesis formula.
    $c_1 = 1$, and it “sits” at ω_1

$$x(t) = 3e^{-j\frac{\pi}{5}} e^{j\omega_1 t}$$

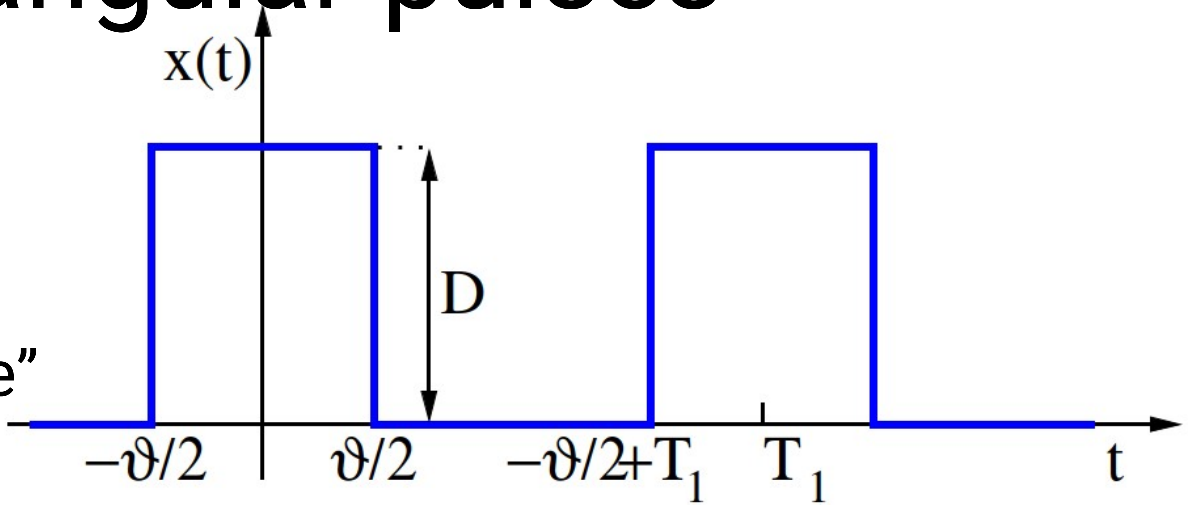
- Complex exponential multiplied b $c_1 = 3e^{-j\frac{\pi}{5}}$ tant:
  - Dtto, the coefficient is the constant
- Examples for $\omega_1 = 2000\pi$ rad/s

#fs_complex_exp

$$x(t) = C_1 \cos(\omega_1 t + \phi_1)$$

- Cosine with a magnitude and phase
- We have seen that we $c_1 = \frac{C_1}{2} e^{j\phi_1}, \; c_{-1} = \frac{C_1}{2} e^{-j\phi_1}$ omplex exponentials at ω_1 and at $-\omega_1$

FS of a series of rectangular pulses



- Period T_1, height D and width of pulses ϑ.
  - The ratio ϑ / T_1 is called “duty cycle” (střída)
- We will need to do some integration here … but first some **intuition**:
  - There are sharp edges, so the spectrum will have lots of high frequencies.
  - Narrow (fast) pulses will lead to even broader spectrum
  - Wide (slow) pulses will lead to narrower spectrum

Preparation I – cardinal sine

- Cardinal sine is defined as $$\mathrm{sinc}(x) = \begin{cases} \frac{\sin(x)}{x} & \text{for} \quad x \neq 0 \\ 1 & \text{for} \quad x = 0 \end{cases}$$
- At zero, the fraction is not defined, so we define “by hand” sinc*(0) = 1*.
- Also called “Mexican hat” function.
- We will soon need to plot this function as **complex numbers** – split into **magnitudes and phases.**

#sinc – beware of difference between numpy’s and my definition !

Preparation II – integral of e^{jxy}

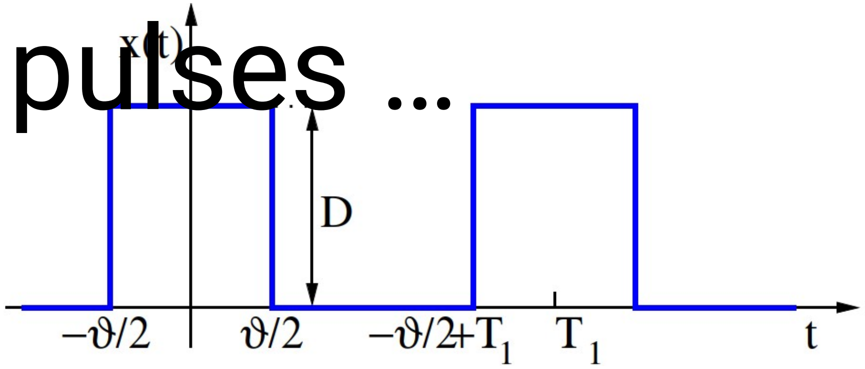
- We will soon need a finite integra $I(x) = \int_{-b}^{b} e^{\pm jxy} dy.$
  - Dubbed "Euler's integral" or "Šebesta's tool" (Šebestova pomůcka - ŠP)
- Some operations (primitive function, usi $\sin\alpha = \frac{e^{j\alpha}-e^{-j\alpha}}{2j}$ and multiplying by *b/b*):

$$I(x) = \left[\frac{e^{\pm jxy}}{\pm jx}\right]_{-b}^{b} = \frac{e^{jxb} - e^{-jxb}}{jx} = 2\frac{e^{jxb} - e^{-jxb}}{2jx} = 2\frac{\sin bx}{x} = 2b\frac{\sin bx}{bx} = 2b \operatorname{sinc} bx$$

- We can verify that it works also for *x = 0*, therefore:

$$I(x) = \left[\frac{e^{\pm jxy}}{\pm jx}\right]_{-b}^{b} = 2b \operatorname{sinc} bx$$

Now finally ready for FS coefficients of the sequence of rectangular pulses …



- By definition:

$$c_k = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{+\frac{T_1}{2}} x(t)e^{-jk\omega_1 t}dt$$

- But it is enough to integrate from −ϑ/2 to +ϑ/2:

$$c_k = \frac{1}{T_1}\int_{-\frac{\vartheta}{2}}^{+\frac{\vartheta}{2}} De^{-jk\omega_1 t}dt = \frac{D}{T_1}\int_{-\frac{\vartheta}{2}}^{+\frac{\vartheta}{2}} e^{-jk\omega_1 t}dt.$$

- And apply ŠP:

$$I(x) = \left[\frac{e^{\pm jxy}}{\pm jx}\right]_{-b}^{b} = 2b \operatorname{sinc} bx$$

$$c_k = \frac{D}{T_1}2\frac{\vartheta}{2}\operatorname{sinc}\left(\frac{\vartheta}{2}k\omega_1\right) = D\frac{\vartheta}{T_1}\operatorname{sinc}\left(\frac{\vartheta}{2}k\omega_1\right).$$

- Now we could use our favorite mathematical program (Python, Excell) and let it `stem` the coefficients, but let us go step-by-step.

FS coefficients of the sequence of rectangular pulses – step 1

- Let us start with an **auxiliary function** (pomocná funkce) with any angular frequency ω. $D\frac{\vartheta}{T_1}\text{sinc}\left(\frac{\vartheta}{2}\omega\right)$
- This function will intersect axis ω for argumer of the function equal π (and its multiples) $\frac{\vartheta}{2}\omega_x = \pi$
- That is for frequency and its multiple $\omega_x = \frac{2\pi}{\vartheta}$

#fs_rectangles_auxiliary

… note the split of **real** cardinal sine into **magnitudes and phases**

- positive real number *a* has magnitude *a* and phase zero (or, for masochists, any multiple of *2π*).
- Negative real number -*a* has magnitude *a* and phase +*π* or - *π* (or, for masochists, any odd multiple of *π*…).
- We can pick phase *π or −π* voluntarily, but we will try to have positive ones for positive values of x and negatiove ones "on the left"

FS coefficients of the sequence of rectangular pulses – step 2

- When the auxiliary function is ready, we can fill the FS coefficients under it – at multiples of the fundamental angular frequency ω_{1}.
- Note that one coefficient c_k is given by both its magnitude and phase, therefore, we need to look into both plots.

#fs_rectangles_full

An example for f_1 = 1 MHz, i.e. T_1 = 1 μs, $\omega_1 = 2\text{x}10^6\,\pi$ rad/s = 2M π ☺
$\vartheta = T_1 / 2$, so that the duty cycle is 50%. Height D = 6.

#fs_rectangles_1Mhz

Synthesizing sequence of rectangular pulses from FS coefficients

$$x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_1 t}$$

- We take the synthesis formula and keep adding components
- Starting with $k = 0$: the d.c. component
- Then going in pairs of coefficients c_k and c_{-k} to ensure that we are always adding cosines.
- Omitting even coefficients k as the corresponding coefficients c_k are zero.

#fs_rectangles_1Mhz_synthesis

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Simples changes of the signal

- Changing the D.C. offset $y(t) = x(t) - 3$: only c_k changes
- Making the signal negative $y(t) = -x(t)$: all coefficients c_k go negative
  - The magnitude does not change
  - Phase will change from $+\pi$ or $-\pi$ to zero and vice versa.
- Faster signal $y(t) = x(2t)$: the whole spectrum will be wider, but the same values of c_k!
- Narrower pulse $\vartheta = T_1 / 4$, duty cycle 25%, otherwise the same:
  - the auxiliary function will be wider (as smaller ϑ leads to wider lobes (laloky)) of the cardinal sine.
  - The multiples of ω_1 will be the same as before.
  - The coefficients will be smaller (signal "working less").

#fs_rectangles_1Mhz_zero_dc

#fs_rectangles_1Mhz_minus

#fs_rectangles_1Mhz_faster

#fs_rectangles_1Mhz_narrower

Shifts of signal

- In case the signal is delayed $y(t) = x(t\text{-}\tau)$: $$c_{y,k} = \frac{1}{T_1}\int_{T_1} x(t-\tau)e^{-jk\omega_1 t}dt.$$
- We can perform a change of variable $g = t - \tau$ and show that:

$$c_{y,k} = \frac{1}{T_1}\int_{T_1} x(g)e^{-jk\omega_1(g+\tau)}dg \quad = \frac{1}{T_1}\int_{T_1} x(g)e^{-jk\omega_1 g}e^{-jk\omega_1\tau}dg \quad = e^{-jk\omega_1\tau}\frac{1}{T_1}\int_{T_1} x(g)e^{-jk\omega_1 g}dg.$$

- But this is the original coefficient multiplied with a “correction”:

$$c_{y,k} = c_{x,k}e^{-jk\omega_1\tau}$$

- Behavior in magnitudes and phases:
  - Magnitudes do not change: $$|c_{y,k}| = |c_{x,k}||e^{-jk\omega_1\tau}| = |c_{x,k}|1 = |c_{x,k}|$$
  - Phases change by subtracting a linear function with slope $-\tau$

$$\arg c_{y,k} = \arg c_{x,k} + \arg e^{-jk\omega_1\tau} = \arg c_{x,k} - k\omega_1\tau.$$

#fs_rectangles_1Mhz_delayed
#fs_rectangles_1Mhz_advanced

Power from FS – Parseval's theorem

- Remember that the power of periodic signal can be obtained:

$$P_s = \frac{1}{T_1}\int_{T_1} |x(t)|^2 dt$$

- And we can have the signal represented by the FS synthesis formula:

$$\boxed{x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_1 t}}$$

- We will be able to obtain the power also by summing the powers c

$$P_s = \sum_{k=-\infty}^{+\infty} \frac{1}{T_1}\int_{T_1} |c_k e^{jk\omega_1 t}|^2 dt = \sum_{k=-\infty}^{+\infty} \frac{1}{T_1}\int_{T_1} |c_k|^2 |e^{jk\omega_1 t}|^2 dt = \sum_{k=-\infty}^{+\infty} \frac{1}{T_1}\int_{T_1} |c_k|^2 dt.$$

- ...er T_1 is this constant multiplied by T_1,

$$P_s = \sum_{k=-\infty}^{+\infty} \frac{1}{T_1} T_1 |c_k|^2 dt = \sum_{k=-\infty}^{+\infty} |c_k|^2 dt.$$

Agenda

- Basic information, operations with time
- Important signals – with links to discrete ones
- Periodic signals
- Energy and power
- Spectral analysis of periodic signals – Fourier series
- FS of typical signals
- Properties of FS and power by FS
- **Spectral analysis of non-periodic signals: Fourier transform**
- FT of typical signals
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Fourier transform (Fourierova transformace) – what do we want

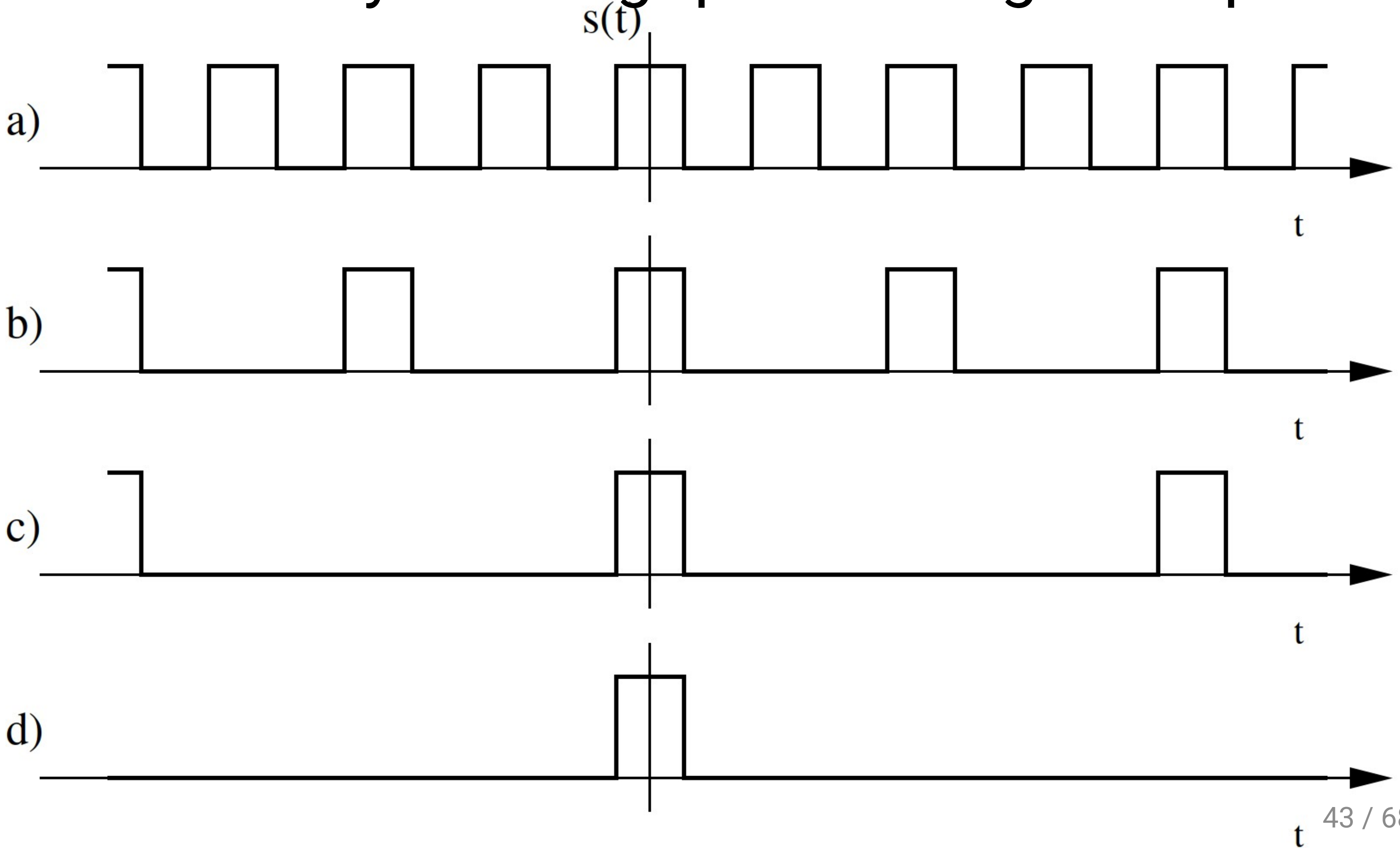
- We want to analyze also non-periodic signals.
- We are asking three usual questions:
  - Where will be the frequencies ? For non-periodic signals, we can not determine fundamental frequency ω_1 and its multiples => we will need to process the whole frequency axis ω
  - The result will not be coefficients, but a **spectral function** (spektrální funkce) $X(j\omega)$.
  - "How much ?" will be answered by the magnitude of spectral function $|X(j\omega)|$
  - "How shifted ?" will be answered by the phase of spectral function $\arg X(j\omega)$.

Definition of FT

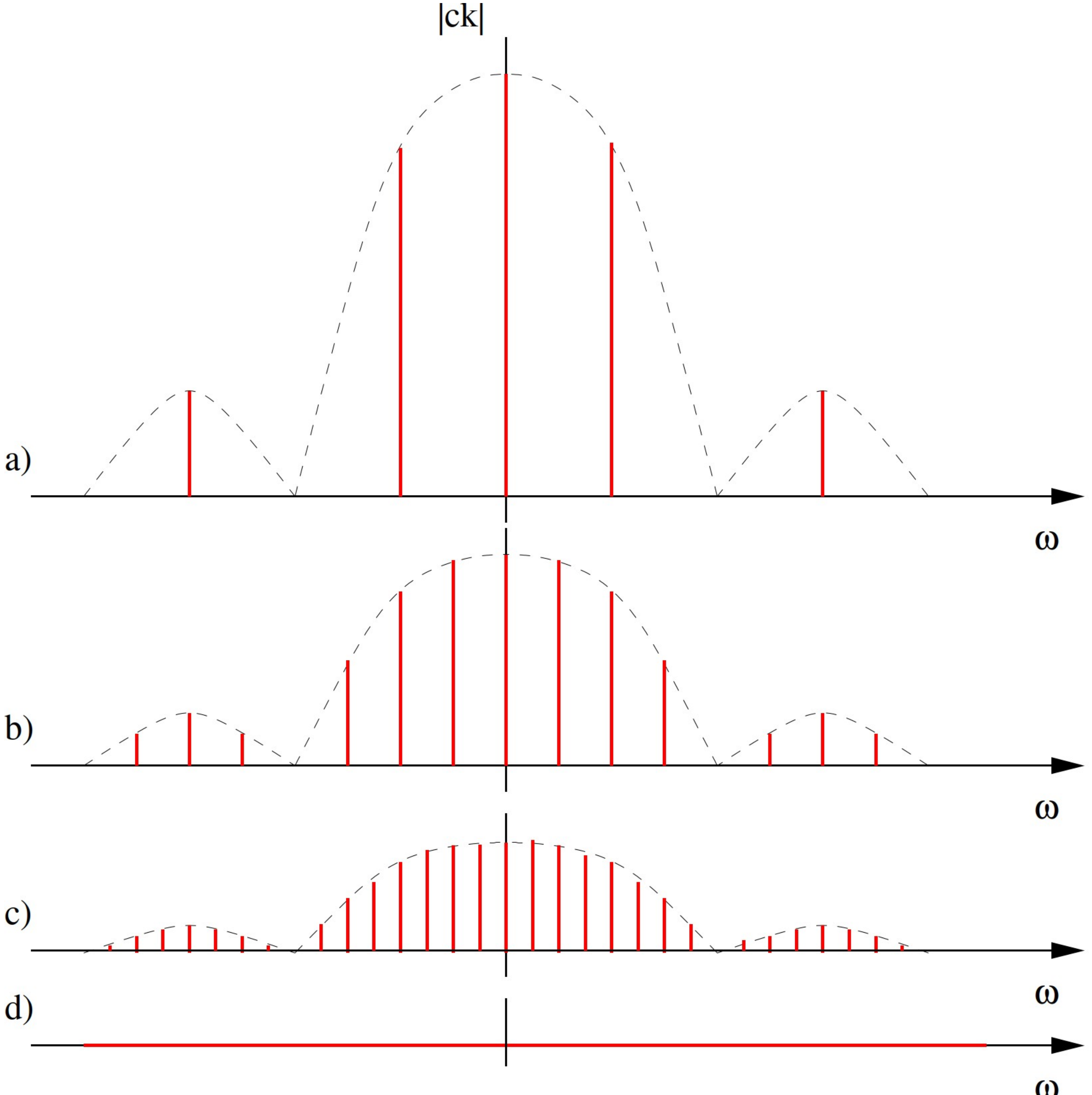
- **Direct FT** (from time to frequency): $$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$
- **Inverse FT** (from frequency to time): $$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{+j\omega t} d\omega.$$
- We can also denote it as:
- For real signals, as usual, v $x(t) \xrightarrow{\mathcal{F}} X(j\omega)$ symm $X(j\omega) \xrightarrow{\mathcal{F}^{-1}} x(t).$ trum:
- As before, we are intrigued by this *j* in notation – why not $X(j\omega) = X^{\star}(-j\omega).$ *X(ω)* but *X(jω)* ?
  - Remember the DTFT case – for discrete signals, we did not denote it *X(ω)* but *X(e^{jω})* !
- As we are getting more experiences in frequency transforms, we note that direct and inverse transform look very similar, except for the **sign**.

How did we get here ?

- Trying to derive FT from FS by “diluting” periodic signals… push T_1 to infinity …



- Does not really work, as the with more and more diluting, the coefficients are smaller and smaller
- for $T_1 = \infty$, they become zero ☹



FS to FT correctly …

- We depart from the definition of Fourier series $c_k = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{+\frac{T_1}{2}} x(t) e^{-jk\omega_1 t} dt$
- We modify for $T_1 \rightarrow \infty$ in a smart way:
  - c_k will become an infinitely small increment of FS coefficient *dc*
  - the limits of integral will become –∞ and +∞.
  - ω_1 will not become zero, but an infinitely small increment $d\omega$
  - with it, we can eliminate the perio $\omega_1 = \frac{2\pi}{T_1} \rightarrow d\omega = \frac{2\pi}{T_1}$ so $\frac{1}{T_1} = \frac{d\omega}{2\pi}$.
  - $k\omega_1$ will become any angular frequency ω
- When done, we obtain: $dc = \frac{d\omega}{2\pi} \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt.$ $2\pi \frac{dc}{d\omega} = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt.$
- But calling the result "infinitely small increment of FS coefficient over infinitely small increment of angular frequ $X(j\omega) = 2\pi \frac{dc}{d\omega}$ would be stupid => **spectral function**.

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FT of Dirac pulse *δ(t)*

- Remember that we defined the sampling property of Dirac pulse:

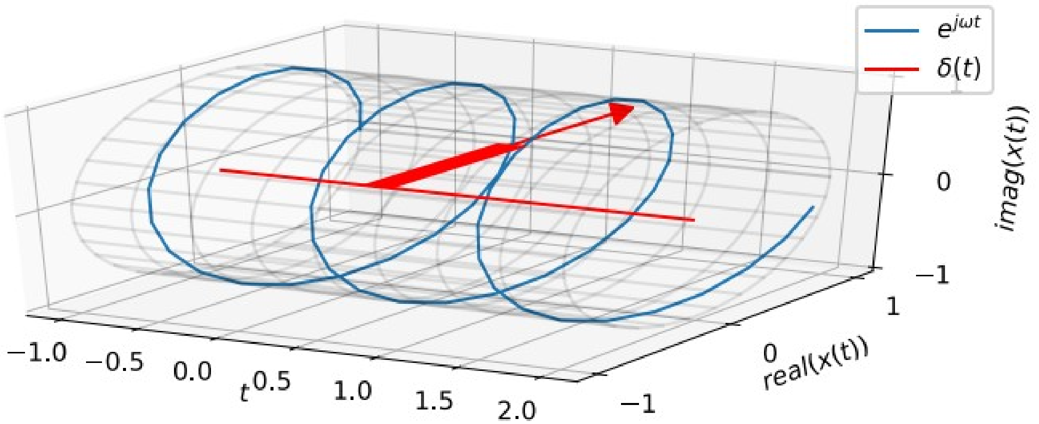
$$\int_{-\infty}^{+\infty} x(t)\delta(t)dt = x(0).$$

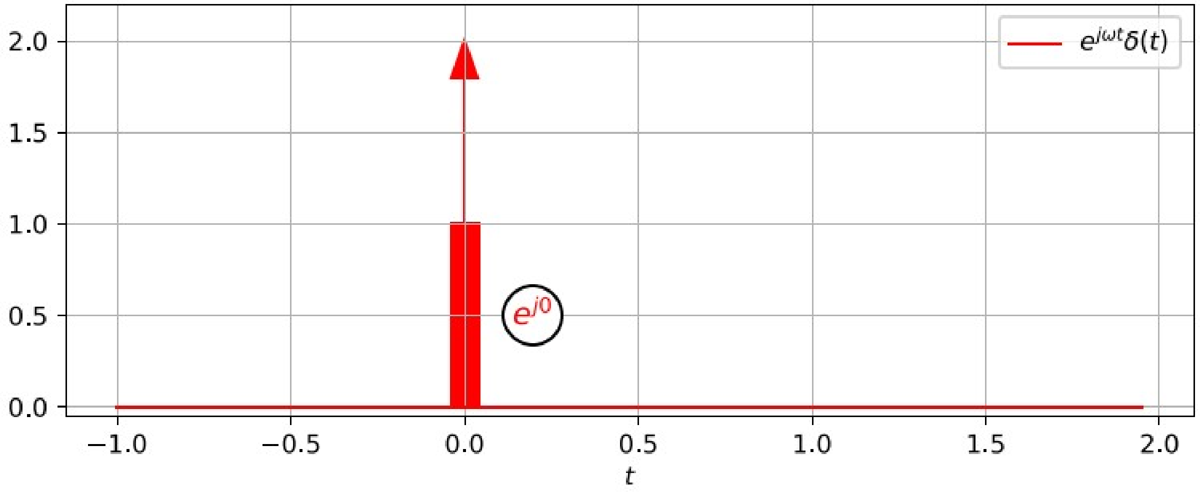
- When standing in the definition of FT, it samples the complex exponential at *t = 0*.

$$X(j\omega) = \int_{-\infty}^{+\infty} \delta(t)e^{-j\omega t}dt.$$

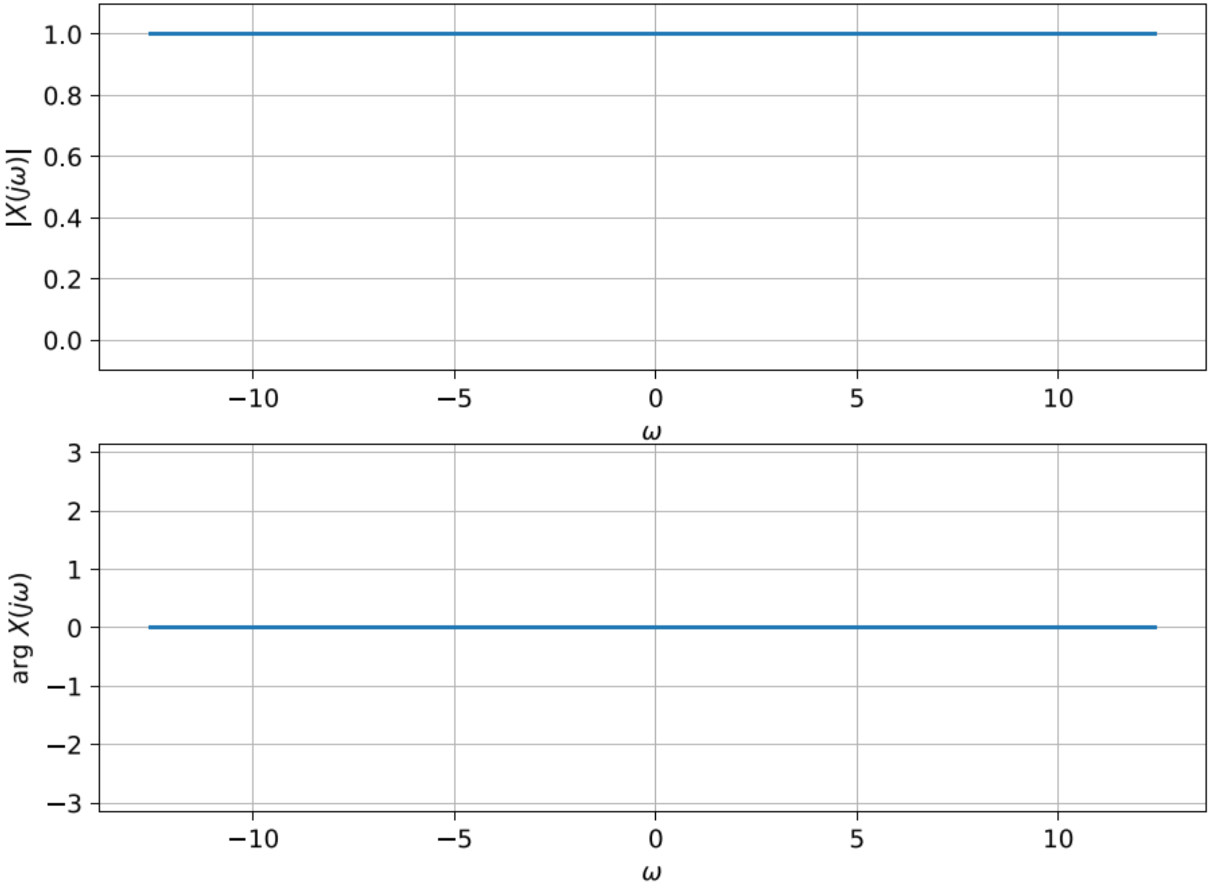
- The integral is thus reduced to

$$X(j\omega) = \int_{-\infty}^{+\infty} \delta(t)e^{j0}dt.$$



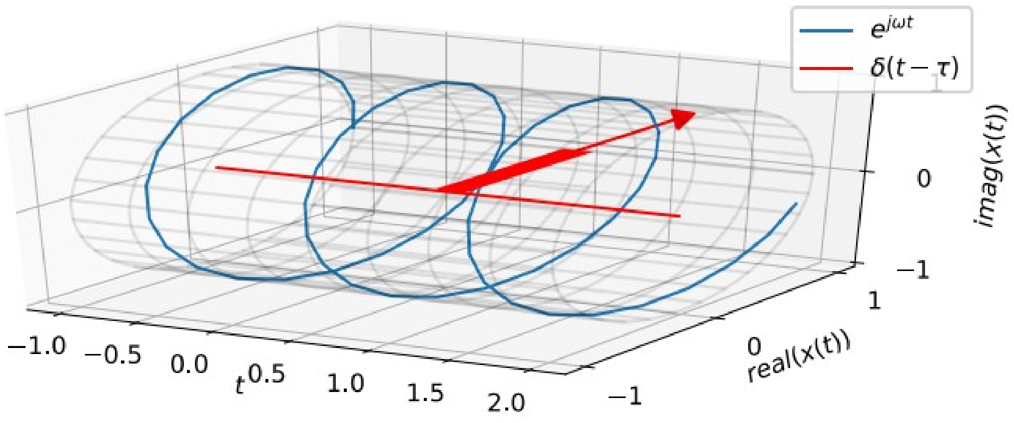


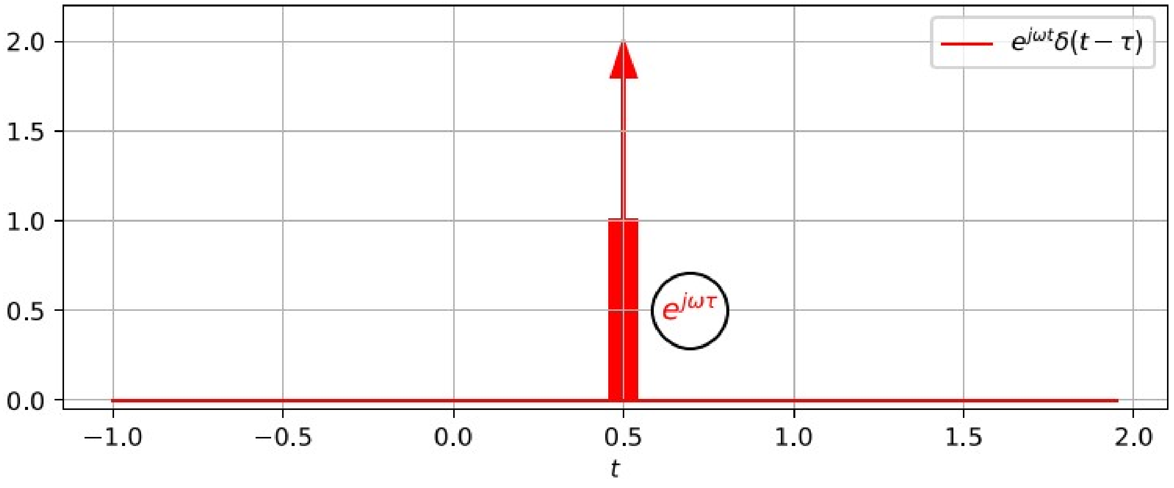
- and the result is *X(jω) = 1*. This is a constant for all frequencies
  - Magnitude 1
  - Phase zero.
- Intuition check: Dirac pulse is a very fast signal, so it should have
  a very wic



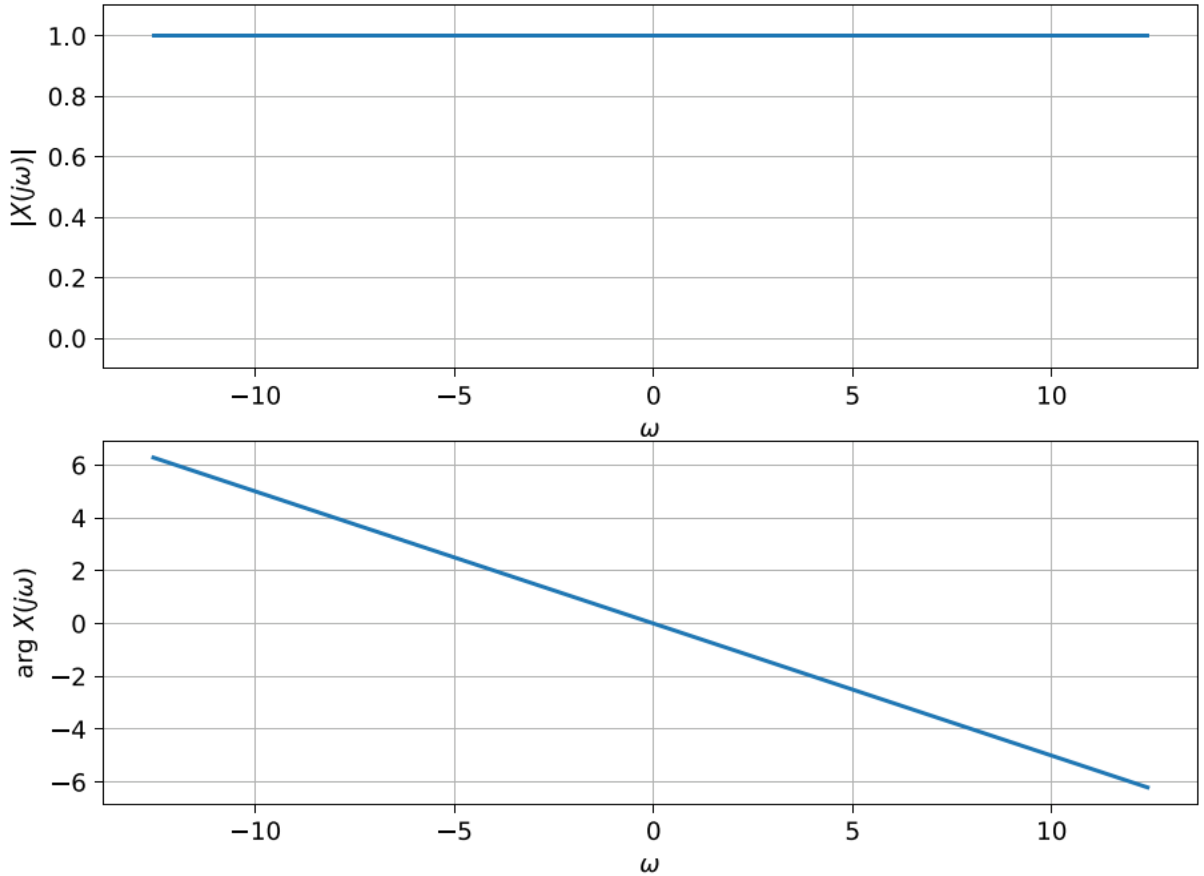
FT of shifted Dirac pulse $\delta(t - \tau)$

- Using again the sampling property $\int_{-\infty}^{+\infty} x(t)\delta(t-\tau)dt = x(\tau).$
- We will work only around time τ: $X(j\omega) = \int_{-\infty}^{+\infty} \delta(t-\tau)e^{-j\omega t}dt.$





- and the result is $X(j\omega) = e^{-j\omega\tau}$. For this expression:
  - Magnitude is 1
  - Phase is linear with slope $-$ $\arg e^{-j\omega\tau} = -\omega\tau$
- Intuition check: shifted Dirac pulse is still a very fast signal, so it should have a very wide spectrum.

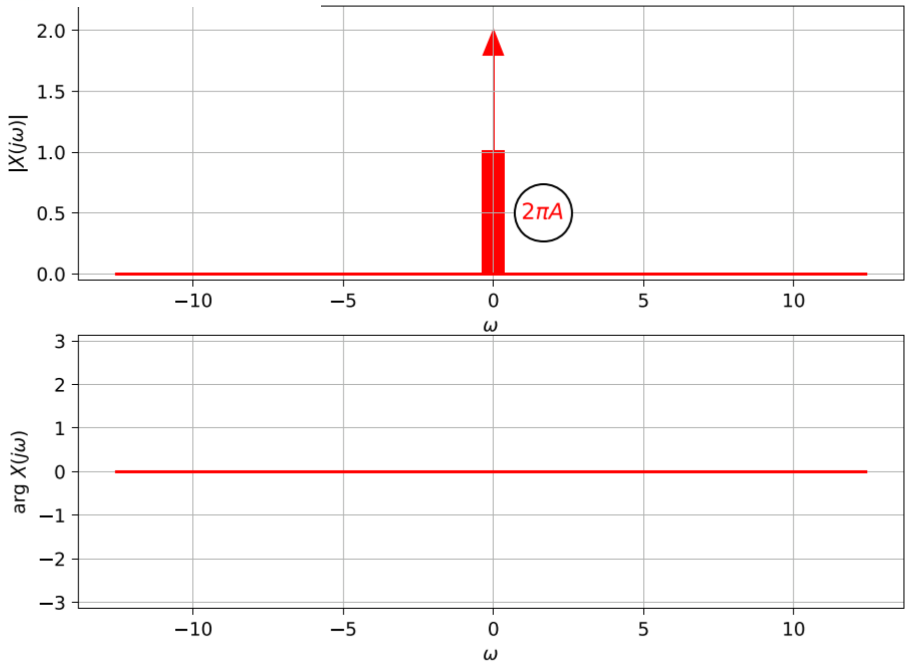


FT of a D.C. signal

- Signal equal to a constant: *x(t) = A*.
- Without derivation, we set *X(jω) = 2πA δ(ω)* and check that it is correct.

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} 2\pi A\delta(\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}2\pi A\int_{-\infty}^{+\infty}\delta(\omega)e^{j\omega t}d\omega = A\int_{-\infty}^{+\infty}\delta(\omega)e^{j\omega t}d\omega.$$

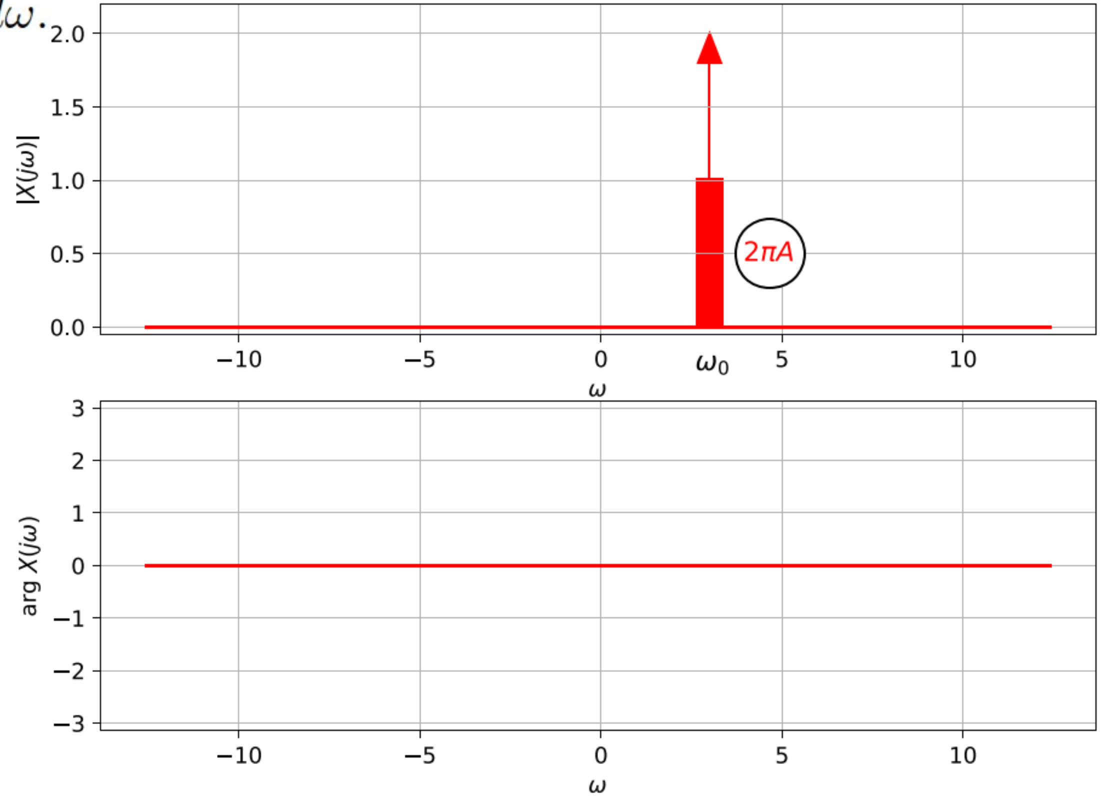
- Dirac pulse has sampling property even in frequency, so the result will be $$x(t) = A\int_{-\infty}^{+\infty} 1\delta(\omega)d\omega,$$
- Therefore, *x(t) = A*.
- Intuition: a very slow signal should have a very narrow spectrum.



FT of a complex exponential

- For $x(t) = e^{j\omega_0 t}$
- We define $X(j\omega) = 2\pi\delta(\omega - \omega_0)$ and we check again …

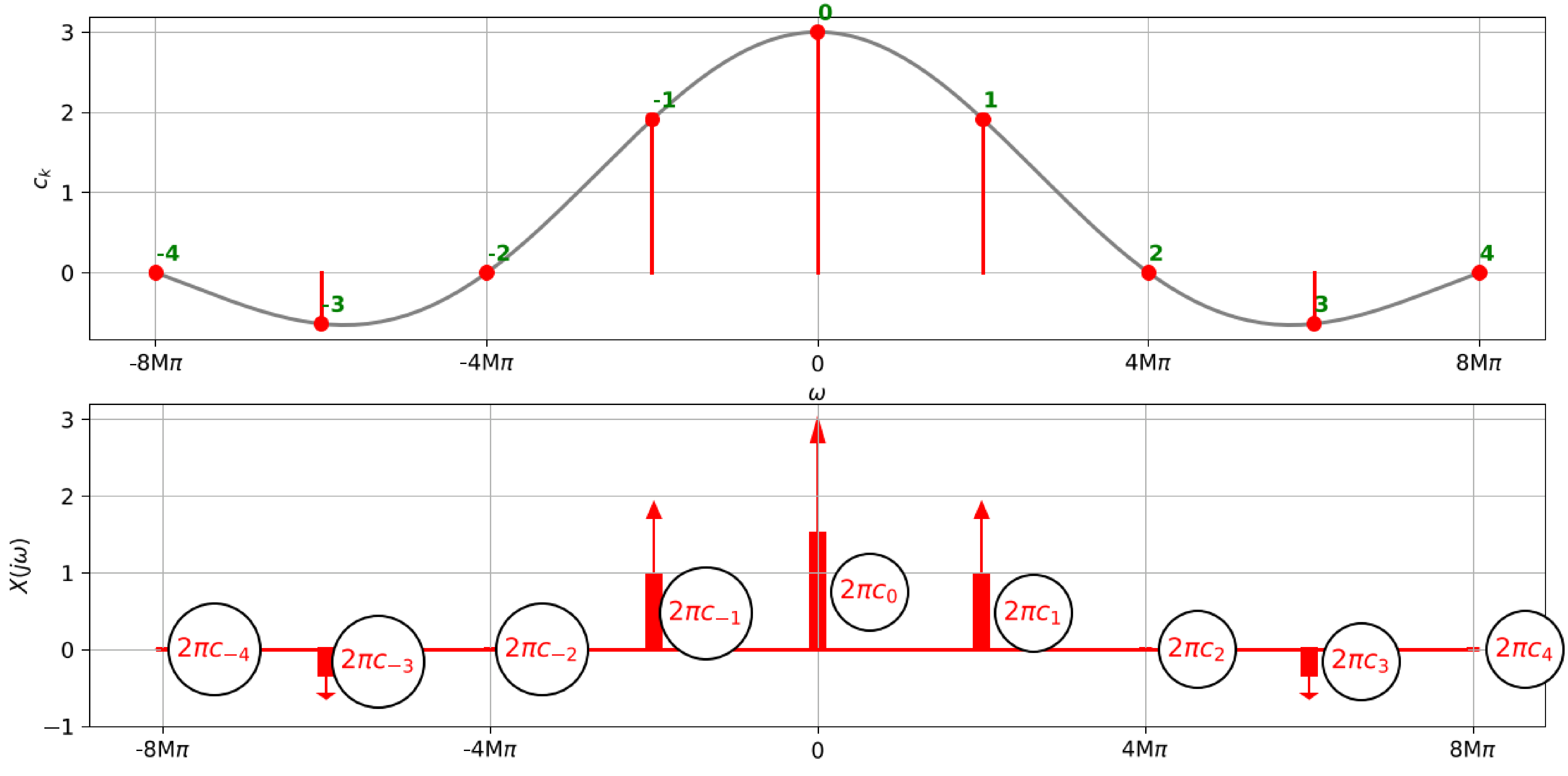
$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} 2\pi\delta(\omega-\omega_0)e^{j\omega t}d\omega = \int_{-\infty}^{+\infty}\delta(\omega-\omega_0)e^{j\omega_0 t}d\omega.$$



- Intuition: a signal that has only one frequency should have a very narrow spectrum only at this frequency.

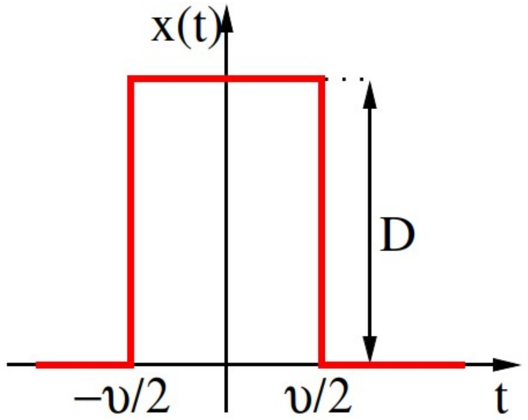
FT of a periodic signal decomposed by FS

- A periodic signal with fundamental period T_1 and angular frequency ω_1 can be decomposed by Fo $x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_1 t}$
- … and we learned that one complex exponential has spectral functio $X(j\omega) = 2\pi\delta(\omega - \omega_0)$
- As all Fourier transforms, FT is linear, so the resulting spectral function will be series of shifted Dirac pulses, each multiplied by 2π and the respective coefficient of $X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi c_k \delta(\omega - k\omega_1)$
- We will be using this trick when deriving the spectral function of a sampled signal.



c_k
ω
-8Mπ
-4Mπ
0
4Mπ
8Mπ
$X(j\omega)$
$2\pi c_{-4}$
$2\pi c_{-3}$
$2\pi c_{-2}$
$2\pi c_{-1}$
$2\pi c_0$
$2\pi c_1$
$2\pi c_2$
$2\pi c_3$
$2\pi c_4$

Rectangular pulse and its FT



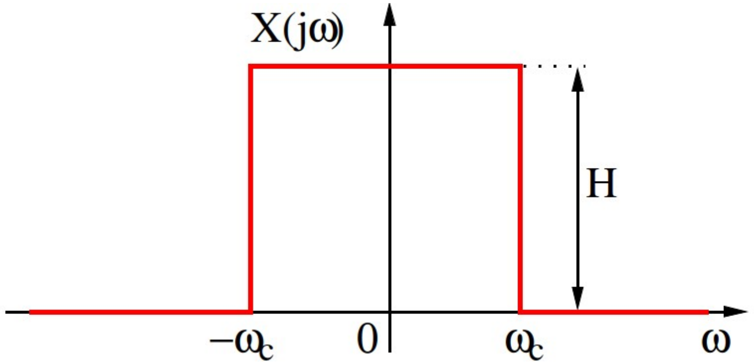
- Similar definition as above, but this time, the pulse is not periodic.
- height *D* and width of pulse ϑ.
- We will again need to do some integration here … but first some **intuition**:
  - There are sharp edges, so the spectrum will have lots of high frequencies.
  - Narrow (fast) pulse will lead to even broader spectrum
  - Wide (slow) pulse wi
- Using FT definition $$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}dt = D\int_{-\frac{\vartheta}{2}}^{+\frac{\vartheta}{2}} e^{-j\omega t}dt.$$

Spectral function of rectangular pulse

- Making use of Šebesta's too $\int_{-b}^{b} e^{\pm jxy} dy = 2b \,\mathrm{sinc}(bx)$ $b = \frac{\vartheta}{2}$, $y = t$ a $x = \omega$
- We obtain $X(j\omega) = D2\frac{\vartheta}{2}\mathrm{sinc}\left(\frac{\vartheta}{2}\omega\right) = D\vartheta\mathrm{sinc}\left(\frac{\vartheta}{2}\omega\right)$
- … cardinal sine again, but this time it is the ultimate result, not an auxiliary function.
- We are again interested, where does sinc touch the frequency axis, we $\frac{\vartheta}{2}\omega_x = \pi$ and $\omega_x = \frac{2\pi}{\vartheta}$ - same as for FS.
- Plotting again as magnitude and phase, and "beautifying" the plot by using +π for positive frequencies and –π for negative ones.

#ft_rectangle

Signal with a square spectral function



- Let’s try an inverse exercise … rectangular spectral function
- Same process as above, but we need to run inverse FT: $$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{+j\omega t}d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} He^{+j\omega t}d\omega = \frac{H}{2\pi}\int_{-\omega_c}^{\omega_c} e^{+j\omega t}d\omega.$$
- Making use of Šebesta’s to($\int_{-b}^{b} e^{\pm jxy}dy = 2b\,\mathrm{sinc}(bx)$ $\quad b = \omega_c,\ y = \omega\ \ x = t.$
- We obtain $$x(t) = \frac{H}{2\pi}2\omega_c \mathrm{sinc}(\omega_c t) = \frac{H\omega_c}{\pi}\mathrm{sinc}(\omega_c t).$$
- The cardinal sine will intersect *t* axis f $\omega_c t_x = \pi,\ \ t_x = \frac{\pi}{\omega_c}.$

#ift_of_rectangular_spectral_function

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Properties of FT

- We will use rectangular pulse with parameters: $\vartheta = 0.5$ μs and $D = 600000$ (to get reasonable values of spectral function).

#ft_rectangle_0.5us

Linearity and inversion

- FT is linear, in case we have independen $x_1(t) \xrightarrow{\mathcal{F}} X_1(j\omega), \quad x_2(t) \xrightarrow{\mathcal{F}} X_2(j\omega)$
- The FT of mix $x(t) = ax_1(t) + bx_2(t)$
- is exactly the same mix of original spectral functi $X(j\omega) = aX_1(j\omega) + bX_2(j\omega).$
- Proof: $X(j\omega) = \int_{-\infty}^{\infty} [ax_1(t) + bx_2(t)]\, e^{-j\omega t} dt \;= a\int_{-\infty}^{\infty} x_1(t)e^{-j\omega t} dt + b\int_{-\infty}^{\infty} x_2(t)e^{-j\omega t} dt.$

#ft_linearity

- For inversion of signa $y(t) = -x(t)$, we obtain the same inversion of spectral function: $Y(j\omega) = -X(j\omega).$
  - The magnitude does not change
  - Phase will change from *+π* or *−π* to zero and vice versa.

#ft_negative

FT of a delayed signal

- In case the signal is delayed *y(t) = x(t-τ)*: $Y(j\omega) = \int_{-\infty}^{\infty} x(t-\tau)e^{-j\omega t}dt.$
- We can perform a change of variable *g = t − τ* and show that:

$$= \int_{-\infty}^{\infty} x(g)e^{-j\omega(g+\tau)}dg, \quad = e^{-j\omega\tau}\int_{-\infty}^{\infty} x(g)e^{-j\omega g}dg.$$

- But this is the original spectral function multiplied with a "correction":

$$Y(j\omega) = e^{-j\omega\tau}X(j\omega).$$

- Let us study the behavior i $|Y(j\omega)| = |e^{-j\omega\tau}||X(j\omega)| = |X(j\omega)|.$ es:
  - Magnitudes do not change:
  - Phases ch $\arg Y(j\omega) = \arg e^{-j\omega\tau} + \arg X(j\omega) = \arg X(j\omega) - \omega\tau.$ -τ:

#ft_rectangles_0.5us_delayed

FT of time-contracted or time-dilated signal

- Modifications of the time axi $y(t) = x(mt),$ speeding up (contraction) for *m > 1*, slowing down (dilation) for *m < 1*
- By definition $Y(j\omega) = \int_{-\infty}^{\infty} x(mt)e^{-j\omega t}dt.$
- Without derivation: $Y(j\omega) = \frac{1}{m}X\left(j\frac{\omega}{m}\right),$
- Interpretation:
  - For *m > 1* (narrower signal), the spectrum should be wider and smaller (less signal to work)
  - For *m < 1* (wider signal), the spectrum should be narrower and bigger (more signal to work)

#ft_rectangles_0.5us_contract_dilate

Energy from FT – Parseval's theorem

- The total energy of signal is given $E_\infty = \int_{-\infty}^{+\infty} |x(t)|^2 dt$ (don't try this for periodic or constant signals …)
- We can try it with one *x(t)* let intact and another given by IFT:

$$\int_{-\infty}^{\infty} x^2(t)dt = \int_{-\infty}^{\infty} x(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \right] dt.$$

- By changing the order of integrations: $= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \underbrace{\left[\int_{-\infty}^{\infty} x(t) e^{j\omega t} dt \right]} d\omega$
- Square bracket is a weird FT, actually the value of spectral function for X(-jω) !
- Knowing that $X(j\omega) = X^\star(-j\omega),$ we can:

$$E_\infty = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) X(-j\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega = \int_{-\infty}^{\infty} L_d(\omega) d\omega.$$

$L_d(\omega) = \frac{|X(j\omega)|^2}{2\pi}$ is called two-sided energy spectral density

#energy_density

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SUMMARY I

- Continuous-time signals are defined for all times … but we can always fake it
  - In plotting by using plot
  - In computing by using numerical integration (sum of values multiplied by time-step)
- In addition to known time operation, we can speed-up or slow-down the time axis.
- Unit step and Dirac pulse are non-existing but very useful signals, Dirac has:
  - zero width, infinite size, surface one
  - very useful for sampling things.
- Periodic signals repeat
  - the smallest repetition time is called fundamental period.
  - and its inverse (both in continuous and in discrete time) is fundamental frequency.
- Energy and power are just sums or integrals of squarred signal values
  - raw for energy
  - normalized by length for power

SUMMARY II

- Spectral analysis of periodic signals: Fourier series (FS)
  - the result are complex coefficients sitting at multiples of fundamental frequency, must be plotted two plots: magnitude and phases.
  - two opposite coefficients are complex conjugate and make up for one cosine.
  - they are computed by projecting the signal on complex exponentials.
- FS of typical signals
  - complex exponential has one FS coefficient.
  - cosine has two FS coefficients.
  - sequence of rectangles has infinity FS coefficients, decaying towards high frequencies according to cardinal sine.
- Properties of FS
  - when the signal is shifted, nothing happens to the magnitudes, phases go downhill (delay) or uphill (advance)
  - in case the signal is contracted (dilated) the values of FS coefficients are the same, but they "sit" at different frequencies.
  - power can be computed from FS coefficients (Parseval).

SUMMARY III

- Spectral analysis of non-periodic signals: Fourier transform (FT)
  - the result is spectral function defined for all frequencies
  - spectral function is computed by projecting the signal on complex exponentials.
- Typical signals
  - Dirac has constant spectral function with flat (Dirac at zero) or inclined phase (Dirac not at zero)
  - spectral function of a D.C. signal is Dirac.
  - spectral function of a complex exponential is shifted Dirac.
  - periodic signal can be broken by FS which can be converted to Diracs at multiples of fundamental frequency.
  - rectangular signal has spectral function given by cardinal sine.
  - dtto in the inverse direction: signal corresponding to square spectral function is cardinal sine.
- Properties of FT
  - FT (and FS) is linear
  - FT of a delayed signal has the same magnitude, phase shifts downhill (delay) or uphill (advance)
  - FT of time contracted signal goes in the opposite way: wider for narrower signal, narrower for wider signal.
  - Energy of a signal can be obtained from its FT (converted to energy spectral density).

Basic truths on signals and spectra

- Spectrum of periodic signal is discrete (coefficients) – Fourier **series**
- Spectrum of non-periodic signal is continuous (function) – Fourier **transform**
- Spectrum of narrow (fast) signal is wide.
- Spectrum of wide (slow) signal is narrow.
- Any sharp edge will result in infinitely long spectrum.
- Spectrum of delayed signal has the same magnitude as before, phase goes downhill.
- Spectrum of advanced signal has the same magnitude as before, phase goes uphill.

Time and spectral domains are like fire and water: always do the opposite things!